Elastic Net

1 Setup and motivation

Take $\boldsymbol{X} \in \mathbb{R}^{N\times(p+1)}$; for a linear regression model, the fitted values are

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}} = \hat{\beta}_0 + x_1\hat{\beta}_1 + \cdots + x_p\hat{\beta}_p.$$

Two criteria are typically used to evaluate the quality of a model:

(i) Accuracy of prediction on "new" data.

(ii) Interpretability (parsimony is important).

Ridge regression can achieve better prediction through the bias–variance tradeoff, but cannot produce a parsimonious model.

The Lasso does both shrinkage and variable selection. The main concern is that if there is a group of variables with high pairwise correlations, then the Lasso tends to select only one variable in that group "at random".

The Elastic Net simultaneously does variable selection and shrinkage, and it can select groups of correlated variables.

Consider data $\{y_i, \boldsymbol{x}_i\}_{i=1}^m$ where $\boldsymbol{x}_i \in \mathbb{R}^p$. Let $\boldsymbol{X} = [\boldsymbol{x}_1 \mid \cdots \mid \boldsymbol{x}_p]$, where $\boldsymbol{x}_j = (x_{1j}, \ldots, x_{mj})'$ for $j = 1, \ldots, p$.

Assume the data is standardized such that

$$\sum_{i=1}^m y_i = 0, \qquad \sum_{i=1}^m x_{ij} = 0, \qquad \sum_{i=1}^m x_{ij}^2 = 1, \quad \forall j = 1, \ldots, p.$$

2 The naive Elastic Net criterion

For all $\lambda_1, \lambda_2 \geq 0$, the naive Elastic Net criterion is

$$L(\lambda_1, \lambda_2, \boldsymbol{\beta}) = \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda_1\|\boldsymbol{\beta}\|_1 + \lambda_2\|\boldsymbol{\beta}\|_2^2. \tag{$*$}$$

The naive Elastic Net estimator $\hat{\boldsymbol{\beta}}$ is

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \{L(\lambda_1, \lambda_2, \boldsymbol{\beta})\}.$$

Let $\alpha = \lambda_2/(\lambda_1 + \lambda_2)$; then solving for $\hat{\boldsymbol{\beta}}$ in $(*)$ is equivalent to

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 \quad \text{s.t.} \quad (1-\alpha)\|\boldsymbol{\beta}\|_1 + \alpha\|\boldsymbol{\beta}\|_2^2 \leq t.$$

Here

$$P_\alpha(\boldsymbol{\beta}) = (1-\alpha)\|\boldsymbol{\beta}\|_1 + \alpha\|\boldsymbol{\beta}\|_2^2$$

is the *elastic net penalty*. When $\alpha = 1$, we obtain $\|\boldsymbol{\beta}\|_2^2$, the ridge penalty. When $\alpha = 0$, we obtain $\|\boldsymbol{\beta}\|_1$, the Lasso penalty. We will only look at $\alpha < 1$.

The ridge term is differentiable everywhere:

$$\partial_{\beta_j} \alpha\|\boldsymbol{\beta}\|_2^2 = \partial_{\beta_j} \alpha \sum_{h=1}^{p} \beta_h^2 = 2\alpha\beta_j.$$

But the Lasso term is singular at zero; indeed,

$$\partial_{\beta_j}\big((1-\alpha)\|\boldsymbol{\beta}\|_1\big) = \partial_{\beta_j}\left((1-\alpha)\sum_{h=1}^{p}|\beta_h|\right) = \begin{cases} 1-\alpha & \text{if } \beta_j \to 0^+ \\ -(1-\alpha) & \text{if } \beta_j \to 0^- \end{cases}$$

Hence $P_\alpha(\boldsymbol{\beta})$ is singular at the origin for $\alpha \in [0,1)$.

3 Convexity

Because the ℓ_1 norm is convex and the ℓ_2 norm is strictly convex,

$$P_\alpha(\boldsymbol{\beta}) = (1-\alpha)\|\boldsymbol{\beta}\|_1 + \alpha\|\boldsymbol{\beta}\|_2^2 \quad \text{is strictly convex for } \alpha \in (0,1].$$

Hence, on $\alpha \in (0,1)$, the elastic net penalty function has the characteristics of both the Lasso and ridge regression.

(Level-set sketch: the ℓ_1 ball is a diamond, the ℓ_2 ball is a disk, and the elastic-net penalty level set for $\alpha \in (0,1)$ is a rounded diamond interpolating between the two.)

4 Equivalence with an augmented Lasso problem

Given λ_1, λ_2 and the data, define

$$\boldsymbol{X}^*_{(m+p)\times p} = \frac{1}{\sqrt{1+\lambda_2}}\begin{pmatrix} \boldsymbol{X}_{m\times p} \\ \sqrt{\lambda_2}\, I_{p\times p}\end{pmatrix}, \qquad \boldsymbol{y}^*_{m+p} = \begin{pmatrix} \boldsymbol{y}_m \\ 0_p \end{pmatrix}.$$

Let $\gamma = \dfrac{\lambda_1}{\sqrt{1+\lambda_2}}$ and $\boldsymbol{\beta}^* = \sqrt{1+\lambda_2}\,\boldsymbol{\beta}$. Then the Lasso on $(\boldsymbol{y}^*, \boldsymbol{X}^*)$ with penalty γ is

$$\min_{\boldsymbol{\beta}^*}\left\{\|\boldsymbol{y}^* - \boldsymbol{X}^*\boldsymbol{\beta}^*\|_2^2 + \gamma\|\boldsymbol{\beta}^*\|_1\right\}.$$

Indeed,

$$\iff \min_{\boldsymbol{\beta}^*}\left\{\left\|\begin{pmatrix} \boldsymbol{y} \\ 0_p\end{pmatrix} - \frac{1}{\sqrt{1+\lambda_2}}\begin{pmatrix} \boldsymbol{X} \\ \sqrt{\lambda_2}\, I_p\end{pmatrix}\sqrt{1+\lambda_2}\,\boldsymbol{\beta}\right\|_2^2 + \frac{\lambda_1}{\sqrt{1+\lambda_2}}\left\|\sqrt{1+\lambda_2}\,\boldsymbol{\beta}\right\|_1\right\}$$

$$\iff \min_{\boldsymbol{\beta}^*}\left\{\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \left\|-\sqrt{\lambda_2}\,\boldsymbol{\beta}\right\|_2^2 + \lambda_1\|\boldsymbol{\beta}\|_1\right\}$$

$$\iff \min_{\boldsymbol{\beta}^*}\left\{\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda_2\|\boldsymbol{\beta}\|_2^2 + \lambda_1\|\boldsymbol{\beta}\|_1\right\}, \quad \text{with argmin } \hat{\boldsymbol{\beta}}^*.$$

As $\hat{\boldsymbol{\beta}}^* = \sqrt{1+\lambda_2}\,\hat{\boldsymbol{\beta}} \Rightarrow \hat{\boldsymbol{\beta}} = \dfrac{1}{\sqrt{1+\lambda_2}}\hat{\boldsymbol{\beta}}^*$. So that the naive Elastic Net with parameters (λ_1, λ_2) is equivalent to the Lasso on $(\boldsymbol{y}^*, \boldsymbol{X}^*)$ with penalty $\gamma = \dfrac{\lambda_1}{\sqrt{1+\lambda_2}}$.

Note that $\boldsymbol{X}^*$ has rank p, meaning that the naive Elastic Net can potentially select all p features, which is not the case for the Lasso.

Similarly, this equivalence allows us to use efficient Lasso algorithms to solve the Elastic Net (e.g. LARS).

5 The orthogonal design case

Consider the orthogonal design case where $\boldsymbol{X}'\boldsymbol{X} = I_p$, s.t.

$$\hat{\boldsymbol{\beta}}^{\text{LS}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} = \boldsymbol{X}'\boldsymbol{y} \equiv \boldsymbol{z}.$$

And

$$\|\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta}\|_2^2 = \|\boldsymbol{y}\|_2^2 - \boldsymbol{y}'\boldsymbol{X}\boldsymbol{\beta} - \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{y} + \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta} = \|\boldsymbol{y}\|_2^2 - 2\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{y} + \boldsymbol{\beta}'\boldsymbol{\beta} = \|\boldsymbol{y}\|_2^2 - 2\sum_{j=1}^p \beta_j z_j + \sum_{j=1}^p \beta_j^2.$$

So that the objective function writes

$$L(\lambda_1, \lambda_2, \boldsymbol{\beta}) = \|\boldsymbol{y}\|_2^2 - 2\sum_{j=1}^p \beta_j z_j + \sum_{j=1}^p \beta_j^2 + \lambda_1 \sum_{j=1}^p |\beta_j| + \lambda_2 \sum_{j=1}^p \beta_j^2$$

$$\propto \sum_{j=1}^p \left(\beta_j^2 - 2z_j\beta_j + \lambda_2\beta_j^2 + \lambda_1|\beta_j|\right).$$

We can then solve for each $\hat{\beta}_j$ separately:

$$\hat{\beta}_j = \arg\min_{\beta_j} \left\{\beta_j^2 - 2z_j\beta_j + \lambda_2\beta_j^2 + \lambda_1|\beta_j|\right\}, \quad \forall j = 1, \ldots, p.$$

(i) $\beta_j > 0$**:** the first-order condition (F.O.C.) is

$$2\hat{\beta}_j - 2z_j + 2\lambda_2\hat{\beta}_j + \lambda_1 = 0 \iff \hat{\beta}_j(1+\lambda_2) = z_j - \lambda_1/2 \iff \hat{\beta}_j = \frac{z_j - \lambda_1/2}{1+\lambda_2} \quad \text{(requiring } z_j > \lambda_1/2\text{)}.$$

(ii) $\beta_j < 0$**:** the F.O.C. leads to

$$\hat{\beta}_j = \frac{z_j + \lambda_1/2}{1+\lambda_2} \quad \text{(requiring } z_j < -\lambda_1/2\text{)}.$$

(iii) $\beta_j = 0$**:** $-2z_j + \lambda_1\partial_{\beta_j}|0| = 0$. As

$$\partial_{\beta_j}|0| \in [-1,1] \implies \lambda_1\partial_{\beta_j}|0| \in [-\lambda_1, \lambda_1] \implies -2z_j + \lambda_1\partial_{\beta_j}|0| \in [-2z_j - \lambda_1,\ -2z_j + \lambda_1].$$

For 0 to be the minimizer we must have $0 \in [-2z_j - \lambda_1, -2z_j + \lambda_1]$

$$\iff -\lambda_1 \le 2z_j \le \lambda_1 \iff |z_j| \le \lambda_1/2.$$

Combining, we obtain

$$\hat{\beta}_j^{\mathrm{NE}} = \begin{cases} \dfrac{z_j - \lambda_1/2}{1+\lambda_2} & \text{if } z_j > \lambda_1/2 \\ \dfrac{z_j + \lambda_1/2}{1+\lambda_2} & \text{if } z_j < -\lambda_1/2 \\ 0 & \text{if } |z_j| \le \lambda_1/2 \end{cases} = \frac{\max\{|z_j| - \lambda_1/2,\, 0\}}{1+\lambda_2}\,\mathrm{sign}(z_j)$$

$$= \frac{\max\left\{|\hat{\beta}_j^{\mathrm{OLS}}| - \lambda_1/2,\, 0\right\}}{1+\lambda_2}\,\mathrm{sign}\left(\hat{\beta}_j^{\mathrm{OLS}}\right), \qquad \forall j = 1, \ldots, p.$$

We know that in that setup we have

$$\hat{\beta}_j^{\mathrm{Ridge}} = \frac{\hat{\beta}_j^{\mathrm{OLS}}}{1+\lambda_2} \qquad \text{and} \qquad \hat{\beta}_j^{\mathrm{Lasso}} = \max\left\{|\hat{\beta}_j^{\mathrm{OLS}}| - \lambda_1/2,\, 0\right\}\mathrm{sign}\left(\hat{\beta}_j^{\mathrm{OLS}}\right).$$

$\Rightarrow$ The naive Elastic Net can be viewed as a two-stage procedure: a ridge-type direct shrinkage followed by a Lasso-type thresholding.

6 A generic penalization framework

Consider the generic penalization method

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda J(\boldsymbol{\beta}), \qquad \text{where } J(\cdot) > 0 \text{ for } \boldsymbol{\beta} \neq \mathbf{0}.$$

The grouping effect

A regression method exhibits the *grouping effect* if the regression coefficients of a group of highly correlated variables tend to be equal.

Suppose that $J(\cdot)$ is strictly convex and $\boldsymbol{x}_i = \boldsymbol{x}_j$ for all $i, j = 1, \ldots, p$.

Suppose $\hat{\beta}_i \neq \hat{\beta}_j$. Define $\tilde{\boldsymbol{\beta}}$ s.t.

$$\hat{\boldsymbol{\beta}} = \begin{pmatrix} \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_i \\ \vdots \\ \hat{\beta}_j \\ \vdots \\ \hat{\beta}_p \end{pmatrix}, \qquad \tilde{\boldsymbol{\beta}} = \begin{pmatrix} \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_j \\ \vdots \\ \hat{\beta}_i \\ \vdots \\ \hat{\beta}_p \end{pmatrix}, \qquad \text{i.e. s.t.} \qquad \tilde{\beta}_i = \hat{\beta}_j, \quad \tilde{\beta}_j = \hat{\beta}_i, \quad \tilde{\beta}_h = \hat{\beta}_h \ \forall h \neq i, j.$$

Take the average $\boldsymbol{\beta}^* = \frac{1}{2}\hat{\boldsymbol{\beta}} + \frac{1}{2}\tilde{\boldsymbol{\beta}}$. Then, for the ith and jth coordinates, we have

$$\beta_i^* = \tfrac{1}{2}\hat{\beta}_i + \tfrac{1}{2}\tilde{\beta}_i = \frac{\hat{\beta}_i + \hat{\beta}_j}{2} = \beta_j^*.$$

Because $\beta_i^* x_i + \beta_j^* x_j = (\beta_i^* + \beta_j^*) x_j = (\hat{\beta}_i + \hat{\beta}_j) x_j$ and all other coefficients are unchanged, we have $\boldsymbol{X}\boldsymbol{\beta}^* = \boldsymbol{X}\hat{\boldsymbol{\beta}}$, so that

$$\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}^*\|_2^2 = \|\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}\|_2^2.$$

Because $\hat{\beta}_i \neq \hat{\beta}_j$, $\hat{\boldsymbol{\beta}} \neq \tilde{\boldsymbol{\beta}}$. Because J is strictly convex,

$$J(\boldsymbol{\beta}^*) = J\left(\frac{\hat{\boldsymbol{\beta}} + \tilde{\boldsymbol{\beta}}}{2}\right) < \tfrac{1}{2}J(\hat{\boldsymbol{\beta}}) + \tfrac{1}{2}J(\tilde{\boldsymbol{\beta}}) = J(\hat{\boldsymbol{\beta}}),$$

where the last equality assumes that J is symmetric (this holds for instance for the elastic net norm).

Then

$$\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}^*\|_2^2 + \lambda J(\boldsymbol{\beta}^*) < \|\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}\|_2^2 + \lambda J(\hat{\boldsymbol{\beta}}),$$

but then $\hat{\boldsymbol{\beta}}$ is not the minimizer, and we must have that $\hat{\beta}_i = \hat{\beta}_j \; \forall i, j$.

This shows that strict convexity of the penalty function guarantees identical coefficients for identical features.

The Lasso does *not* have the grouping effect

Now consider the case where $J(\boldsymbol{\beta}) = \|\boldsymbol{\beta}\|_1$ (the Lasso penalty).

Suppose that $\hat{\beta}_i\hat{\beta}_j < 0$ (i.e. $\text{sign}(\hat{\beta}_i) \neq \text{sign}(\hat{\beta}_j)$).

Define $\tilde{\boldsymbol{\beta}}$ s.t. $\tilde{\beta}_i = \hat{\beta}_i + \hat{\beta}_j$ and $\tilde{\beta}_j = 0$, $\tilde{\beta}_h = \hat{\beta}_h \; \forall h \neq i, j$.

Then $\hat{\beta}_i x_i + \hat{\beta}_j x_j = (\hat{\beta}_i + \hat{\beta}_j)x_j = (\tilde{\beta}_i + \tilde{\beta}_j)x_j$, hence

$$\|\boldsymbol{y} - \boldsymbol{X}\tilde{\boldsymbol{\beta}}\|_2^2 = \|\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}\|_2^2.$$

Because $\hat{\beta}_i$ and $\hat{\beta}_j$ have opposite signs, $|\hat{\beta}_i + \hat{\beta}_j| < |\hat{\beta}_i| + |\hat{\beta}_j|$, thus

$$\|\tilde{\boldsymbol{\beta}}\|_1 = |\hat{\beta}_i + \hat{\beta}_j| + |0| + \sum_{h \neq i,j} |\hat{\beta}_h| < |\hat{\beta}_i| + |\hat{\beta}_j| + \sum_{h \neq i,j} |\hat{\beta}_h| = \|\hat{\boldsymbol{\beta}}\|_1,$$

hence $J(\tilde{\boldsymbol{\beta}}) < J(\hat{\boldsymbol{\beta}})$, contradicting the fact that $\hat{\boldsymbol{\beta}}$ is the minimizer. Therefore we conclude that $\hat{\beta}_i\hat{\beta}_j \geq 0$, i.e. $\text{sign}(\hat{\beta}_i) = \text{sign}(\hat{\beta}_j)$.

Because $\hat{\beta}_i\hat{\beta}_j \geq 0$, we get that $|\hat{\beta}_i + \hat{\beta}_j| = |\hat{\beta}_i| + |\hat{\beta}_j|$.

Consider again $\boldsymbol{\beta}^*$; we have $\beta_i^* + \beta_j^* = \hat{\beta}_i + \hat{\beta}_j$, hence

$$|\beta_i^*| + |\beta_j^*| = |\beta_i^* + \beta_j^*| = |\hat{\beta}_i + \hat{\beta}_j| = |\hat{\beta}_i| + |\hat{\beta}_j|.$$

So that $\|\boldsymbol{\beta}^*\|_1 = \|\hat{\boldsymbol{\beta}}\|_1$, hence $\boldsymbol{X}\boldsymbol{\beta}^* = \boldsymbol{X}\hat{\boldsymbol{\beta}}$.

In general, any convex combination of the two coefficients is also a minimizer. That is, the Lasso exhibits *no grouping effect*: it arbitrarily picks one variable from a group of identical predictors.

7 Quantifying the grouping effect of the naive Elastic Net

Let $\hat{\boldsymbol{\beta}}(\lambda_1, \lambda_2)$ be the naive Elastic Net estimate.

Suppose $\hat{\beta}_i(\lambda_1, \lambda_2)\,\hat{\beta}_j(\lambda_1, \lambda_2) > 0$. Then

$$\text{sign}\Big(\hat{\beta}_i(\lambda_1, \lambda_2)\Big) = \text{sign}\Big(\hat{\beta}_j(\lambda_1, \lambda_2)\Big), \qquad \hat{\beta}_i, \hat{\beta}_j \neq 0.$$

By definition of $\hat{\boldsymbol{\beta}}(\lambda_1, \lambda_2)$, we have

$$\partial_{\beta_h} L(\lambda_1, \lambda_2, \boldsymbol{\beta})\Big|_{\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}(\lambda_1, \lambda_2)} = 0, \quad \text{if } \hat{\beta}_h(\lambda_1, \lambda_2) \neq 0.$$

I.e.

$$\underbrace{-2\boldsymbol{x}_i'\left(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}(\lambda_1,\lambda_2)\right)}_{\hat{\boldsymbol{\varepsilon}}(\lambda_1,\lambda_2)} + \lambda_1 \operatorname{sign}\left(\hat{\beta}_i(\lambda_1,\lambda_2)\right) + 2\lambda_2\hat{\beta}_i(\lambda_1,\lambda_2) = 0,$$

$$-2\boldsymbol{x}_j'\hat{\boldsymbol{\varepsilon}} + \lambda_1 \operatorname{sign}\left(\hat{\beta}_j(\lambda_1,\lambda_2)\right) + 2\lambda_2\hat{\beta}_j(\lambda_1,\lambda_2) = 0.$$

Subtracting,

$$\iff 2\lambda_2\left(\hat{\beta}_i(\lambda_1,\lambda_2) - \hat{\beta}_j(\lambda_1,\lambda_2)\right) - 2(\boldsymbol{x}_i - \boldsymbol{x}_j)'\hat{\boldsymbol{\varepsilon}}(\lambda_1,\lambda_2) = 0$$

(as $\operatorname{sign}(\hat{\beta}_i(\lambda_1,\lambda_2)) = \operatorname{sign}(\hat{\beta}_j(\lambda_1,\lambda_2))$), hence

$$\iff \hat{\beta}_i(\lambda_1,\lambda_2) - \hat{\beta}_j(\lambda_1,\lambda_2) = \frac{1}{\lambda_2}(\boldsymbol{x}_i - \boldsymbol{x}_j)'\hat{\boldsymbol{\varepsilon}}(\lambda_1,\lambda_2).$$

Since the data is standardized: $\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2 = \|\boldsymbol{x}_i\|_2^2 + \|\boldsymbol{x}_j\|_2^2 - 2\boldsymbol{x}_i'\boldsymbol{x}_j = 2(1-\rho)$,

$$\left|\hat{\beta}_i(\lambda_1,\lambda_2) - \hat{\beta}_j(\lambda_1,\lambda_2)\right| = \frac{1}{\lambda_2}\left|(\boldsymbol{x}_i - \boldsymbol{x}_j)'\hat{\boldsymbol{\varepsilon}}(\lambda_1,\lambda_2)\right| \leq \frac{1}{\lambda_2}\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2\,\|\hat{\boldsymbol{\varepsilon}}(\lambda_1,\lambda_2)\|_2 \quad \text{(Cauchy–Schwarz)}$$

$$= \frac{1}{\lambda_2}\sqrt{2(1-\rho)}\,\|\hat{\boldsymbol{\varepsilon}}(\lambda_1,\lambda_2)\|_2.$$

By definition of $\hat{\boldsymbol{\beta}}$:

$$\|\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}\|_2^2 + \lambda_1\|\hat{\boldsymbol{\beta}}\|_1 + \lambda_2\|\hat{\boldsymbol{\beta}}\|_2^2 \leq \|\boldsymbol{y} - \boldsymbol{X}\cdot 0\|_2^2$$

$$\iff \|\hat{\boldsymbol{\varepsilon}}(\lambda_1,\lambda_2)\|_2^2 + \lambda_1\|\hat{\boldsymbol{\beta}}\|_1 + \lambda_2\|\hat{\boldsymbol{\beta}}\|_2^2 \leq \|\boldsymbol{y}\|_2^2 \implies \|\hat{\boldsymbol{\varepsilon}}(\lambda_1,\lambda_2)\|_2^2 \leq \|\boldsymbol{y}\|_2^2$$

$$\implies \|\hat{\boldsymbol{\varepsilon}}(\lambda_1,\lambda_2)\|_2 \leq \|\boldsymbol{y}\|_2 \leq \|\boldsymbol{y}\|_1, \quad \text{hence}$$

$$\left|\hat{\beta}_i(\lambda_1,\lambda_2) - \hat{\beta}_j(\lambda_1,\lambda_2)\right| \leq \frac{1}{\lambda_2}\sqrt{2(1-\rho)}\,\|\boldsymbol{y}\|_1$$

$$\implies D_{\lambda_1,\lambda_2}(i,j) \equiv \frac{\left|\hat{\beta}_i(\lambda_1,\lambda_2) - \hat{\beta}_j(\lambda_1,\lambda_2)\right|}{\|\boldsymbol{y}\|_1} \leq \frac{1}{\lambda_2}\sqrt{2(1-\rho)}. \qquad (**)$$

Hence, if say $\rho = 1$, then the difference between the coefficient paths of predictors i and j is (almost) zero.

The upper bound in inequality $(**)$ provides a quantitative description of the grouping effect of the naive Elastic Net.

8 A Bayesian interpretation

The naive Elastic Net is the MAP estimate under a specific prior distribution. Consider the linear Gaussian regression model:

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, \sigma^2 I_N),$$

the likelihood being

$$p(\boldsymbol{y} \mid \boldsymbol{\beta}, \sigma^2) \propto \exp\left(-\frac{1}{2\sigma^2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2\right),$$

and consider the prior distribution on $\boldsymbol{\beta}$:

$$p(\boldsymbol{\beta} \mid \lambda, \alpha) \propto \exp\left(-\lambda\left\{\alpha\|\boldsymbol{\beta}\|_2^2 + (1-\alpha)\|\boldsymbol{\beta}\|_1\right\}\right), \qquad \lambda > 0,$$

where $\alpha = \dfrac{\lambda_2}{\lambda_1+\lambda_2}$.

Applying Bayes' theorem,

$$p(\boldsymbol{\beta} \mid \boldsymbol{y}, \boldsymbol{X}, \lambda, \alpha) = \frac{p(\boldsymbol{y} \mid \boldsymbol{\beta}, \sigma^2)\, p(\boldsymbol{\beta} \mid \lambda, \alpha)}{p(\boldsymbol{y} \mid \boldsymbol{\beta}, \sigma^2)} \propto p(\boldsymbol{y} \mid \boldsymbol{\beta}, \sigma^2)\, p(\boldsymbol{\beta} \mid \lambda, \alpha)$$

$$= \exp\left(-\frac{1}{2\sigma^2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 - \lambda\left\{\alpha\|\boldsymbol{\beta}\|_2^2 + (1-\alpha)\|\boldsymbol{\beta}\|_1\right\}\right)$$

$$\Longrightarrow -\log p(\boldsymbol{\beta} \mid \boldsymbol{y}, \boldsymbol{X}, \lambda, \alpha) = \frac{1}{2\sigma^2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda\alpha\|\boldsymbol{\beta}\|_2^2 + \lambda(1-\alpha)\|\boldsymbol{\beta}\|_1$$

$$\propto \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda_1\|\boldsymbol{\beta}\|_1 + \lambda_2\|\boldsymbol{\beta}\|_2^2, \qquad \text{where } \lambda_1 = \lambda\alpha,\ \lambda_2 = \lambda(1-\alpha).$$

Hence the MAP estimator of $\boldsymbol{\beta}$ under this prior is exactly the naive Elastic Net estimator.

Note that for $\alpha = 0$ we get the Laplace prior, whose MAP is the Lasso. For $\alpha = 1$, we get the Gaussian prior, whose MAP is Ridge.

9 Correcting the double shrinkage

Because the naive Elastic Net estimator is a 2-stage procedure (ridge then Lasso), it appears to induce a double amount of shrinkage which does not help to reduce the variance much and introduces unnecessary extra bias. We therefore need to correct this double shrinkage.

Consider again the augmented data

$$\boldsymbol{X}^* \equiv \boldsymbol{X}^*_{(m+p)\times p} = \begin{pmatrix} \boldsymbol{X}_{m\times p} \\ \sqrt{\lambda_2}\, I_p \end{pmatrix} \frac{1}{\sqrt{1+\lambda_2}}, \qquad \boldsymbol{y}^* = \begin{pmatrix} \boldsymbol{y}_m \\ 0_p \end{pmatrix}.$$

The naive Elastic Net solves, as we have seen,

$$\hat{\boldsymbol{\beta}}^* = \arg\min_{\boldsymbol{\beta}^*} \|\boldsymbol{y}^* - \boldsymbol{X}^*\boldsymbol{\beta}^*\|_2^2 + \frac{\lambda_1}{\sqrt{1+\lambda_2}}\|\boldsymbol{\beta}^*\|_1.$$

The Elastic Net estimates are *defined* by

$$\hat{\boldsymbol{\beta}}^{\mathrm{EN}} = \sqrt{1+\lambda_2}\, \hat{\boldsymbol{\beta}}^*.$$

Recall that $\hat{\boldsymbol{\beta}}(\text{naive elastic net}) = \dfrac{1}{\sqrt{1+\lambda_2}}\hat{\boldsymbol{\beta}}^*$, thus

$$\hat{\boldsymbol{\beta}}^{\mathrm{EN}} = (1+\lambda_2)\, \hat{\boldsymbol{\beta}}(\text{naive elastic net}).$$

Justifying the $(1+\lambda_2)$ scaling factor

The $(1+\lambda_2)$ scaling factor can be justified. Since $\boldsymbol{X}$ is standardized,

$$\boldsymbol{X}'\boldsymbol{X} = \begin{pmatrix} 1 & \rho_{12} & \cdots & \rho_{1p} \\ & 1 & & \vdots \\ & & \ddots & \rho_{p-1,p} \\ & & & 1 \end{pmatrix}, \qquad \text{recall that}$$

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = (\boldsymbol{X}'\boldsymbol{X} + \lambda_2 I_p)^{-1}\boldsymbol{X}'\boldsymbol{y} \equiv R\boldsymbol{y}.$$

The matrix R can be rewritten as

$$R = (\boldsymbol{X}'\boldsymbol{X} + \lambda_2 I_p)^{-1}\boldsymbol{X}' = \left(\hat{\Sigma} + \tfrac{1}{2} I_p\right)^{-1}\boldsymbol{X}', \qquad \text{where } \hat{\Sigma} = \boldsymbol{X}'\boldsymbol{X},$$

$$= (1+\lambda_2)^{-1}\left(\frac{\hat{\Sigma} + \lambda_2 I_p}{1+\lambda_2}\right)^{-1}\boldsymbol{X}' = \frac{1}{1+\lambda_2}M^{-1}\boldsymbol{X}', \qquad \text{where } M = \frac{\hat{\Sigma}+\lambda_2 I_p}{1+\lambda_2},$$

i.e. where

$$M = \begin{pmatrix} 1 & \frac{\rho_{12}}{1+\lambda_2} & \cdots & \frac{\rho_{1p}}{1+\lambda_2} \\ \frac{\rho_{21}}{1+\lambda_2} & 1 & \cdots & \frac{\rho_{2p}}{1+\lambda_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\rho_{p1}}{1+\lambda_2} & \frac{\rho_{p2}}{1+\lambda_2} & \cdots & 1 \end{pmatrix}, \qquad \text{hence}$$

$$R = \frac{1}{1+\lambda_2}R^* = \frac{1}{1+\lambda_2}\begin{pmatrix} 1 & \frac{\rho_{12}}{1+\lambda_2} & \cdots & \frac{\rho_{1p}}{1+\lambda_2} \\ & 1 & \cdots & \vdots \\ & & \ddots & \frac{\rho_{p-1,p}}{1+\lambda_2} \\ & & & 1 \end{pmatrix}^{-1}\boldsymbol{X}'.$$

From the above expression, we can interpret the ridge operator as decorrelation followed by direct scaling.

Elastic Net as a rescaled Lasso

Denote $\hat{\boldsymbol{\beta}}^{\text{EN}} = \hat{\boldsymbol{\beta}}$; we explained that it is defined by $\hat{\boldsymbol{\beta}} = \sqrt{1+\lambda_2}\,\boldsymbol{\beta}^*$, substituting into the augmented Lasso objective:

$$\left\|\boldsymbol{y}^* - \boldsymbol{X}^*\frac{\boldsymbol{\beta}}{\sqrt{1+\lambda_2}}\right\|_2^2 + \gamma\left\|\frac{\boldsymbol{\beta}}{\sqrt{1+\lambda_2}}\right\|_1 = a + b,$$

where

$$a = \|\boldsymbol{y}^*\|_2^2 + \left(\frac{1}{\sqrt{1+\lambda_2}}\right)^2\|\boldsymbol{X}^*\boldsymbol{\beta}\|_2^2 - 2\boldsymbol{y}^{*\prime}\boldsymbol{X}^*\frac{\boldsymbol{\beta}}{\left(\sqrt{1+\lambda_2}\right)^2}, \qquad \text{and:}$$

$$\|\boldsymbol{y}^*\|_2^2 = \boldsymbol{y}^{*\prime}\boldsymbol{y}^* = \begin{pmatrix}\boldsymbol{y}_m' & 0_p'\end{pmatrix}\begin{pmatrix}\boldsymbol{y}_m \\ 0_p\end{pmatrix} = \boldsymbol{y}'\boldsymbol{y},$$

$$\boldsymbol{y}^{*\prime}\boldsymbol{X}^* = \begin{pmatrix}\boldsymbol{y}' & 0_p'\end{pmatrix}\begin{pmatrix}\boldsymbol{X} \\ \sqrt{\lambda_2}\,I_p\end{pmatrix}\frac{1}{\sqrt{1+\lambda_2}} = \frac{\boldsymbol{y}'\boldsymbol{X}}{\sqrt{1+\lambda_2}}, \qquad \text{so that:}$$

$$a = \boldsymbol{y}'\boldsymbol{y},$$

$$\boldsymbol{X}^{*\prime}\boldsymbol{X}^{*} = \begin{pmatrix}\boldsymbol{X}' & \sqrt{\lambda_2}\, I_p\end{pmatrix}\begin{pmatrix}\boldsymbol{X} \\ \sqrt{\lambda_2}\, I_p\end{pmatrix}\frac{1}{1+\lambda_2} = \frac{\boldsymbol{X}'\boldsymbol{X} + \lambda_2 I_p}{1+\lambda_2}, \qquad \text{so that:}$$

$$a = \boldsymbol{y}'\boldsymbol{y} + \frac{1}{1+\lambda_2}\boldsymbol{\beta}'(\boldsymbol{X}'\boldsymbol{X} + \lambda_2 I_p)\boldsymbol{\beta} - 2\,\frac{\boldsymbol{y}'\boldsymbol{X}}{\sqrt{1+\lambda_2}}\,\frac{\boldsymbol{\beta}}{\sqrt{1+\lambda_2}}.$$

So that the objective function becomes

$$\propto \boldsymbol{\beta}'\frac{\boldsymbol{X}'\boldsymbol{X} + \lambda_2 I_p}{(1+\lambda_2)^2}\boldsymbol{\beta} - 2\boldsymbol{y}'\boldsymbol{X}\frac{\boldsymbol{\beta}}{1+\lambda_2} + \frac{\gamma}{\sqrt{1+\lambda_2}}\|\boldsymbol{\beta}\|_1$$

$$\propto \boldsymbol{\beta}'\frac{\boldsymbol{X}'\boldsymbol{X} + \lambda_2 I_p}{1+\lambda_2}\boldsymbol{\beta} - 2\boldsymbol{y}'\boldsymbol{X}\boldsymbol{\beta} + \lambda_1\|\boldsymbol{\beta}\|_1.$$

Minimizing this objective is equivalent to the Lasso, but replacing the sample correlation matrix $\boldsymbol{X}'\boldsymbol{X}$ with $(\boldsymbol{X}'\boldsymbol{X} + \lambda_2 I_p)/(1+\lambda_2)$.

Where

$$\frac{\boldsymbol{X}'\boldsymbol{X} + \lambda_2 I_p}{1+\lambda_2} = (1-\xi)\hat{\Sigma} + \xi I_p, \qquad \text{where } \xi = \frac{\lambda_2}{1+\lambda_2}.$$

Hence, rescaling after Elastic Net penalization is equivalent to replacing $\hat{\Sigma}$ with its shrunken version in the Lasso.

If only training data are available, cross-validation (CV) can be used to estimate prediction error and compare different models (λ_1, λ_2).